

or tools that can be connected to third party IDEs. There are a couple of platforms that offer a combination of all options.

4. **Security:** While considering the security features of platforms, developers must look for compliance with industry standards, data encryption capabilities and systems monitoring. All the leading vendors such as Microsoft, Sencha and PhoneGap comply with industry standards and provide multi-level security for mobile applications. A thorough review of the prospective development platform should be made in order to ensure that it meets compliance and security needs. Apart from all this, different problems that might arise while integrating the mobile application development platform with other security and enterprise systems should be considered.
5. **Video or audio processing:** Video/audio channels are perhaps the most whimsical feature of any mobile software application. Only platforms that support high quality video processing, which ensure that apps behave their best, should be considered.
6. **Usability:** It is important for any mobile app development platform's interface to be user-friendly and highly intuitive. It should work the way the application developers expect it to and should have the ability to be enhanced to enable collaboration in case a development team needs to work concurrently during the build cycle.
7. **Integration with ERP and CRM:** It's always good to choose a mobile application development platform that can easily integrate with CRM or ERP, which are the key requirements of a mobile application.
8. **Costs:** The total cost associated with different licensing development tools, the required infrastructure, development cycles, and deployment of any mobile app can really become worryingly high. However, there are a couple of cloud based services which offer options that can take care of most of the workload. This can reduce the total cost involved in developing a mobile application.
9. **Skills required to use the platform:** Even if an enterprise decides to choose the low-code path, some development code needs to be written, eventually. Hence,

skill sets or programming language skills required to use the platform chosen should be sought so that the platform can be effectively utilised. If any business is in a hurry to get the mobile app to the market, developers might opt for services that offer code samples or templates.

Let's now have a quick look at some of the widely used mobile app development platforms.

PhoneGap

PhoneGap is an open source program for developing apps for Android, iOS and Windows mobile devices. It makes use of standard Web development programming languages like HTML, CSS and JavaScript. This helps mobile application developers to make use of different device hardware features like accelerometers, cameras, sound, GPS location, etc. It also offers an Adobe AIR application and online training courses, to help access native APIs and build mobile apps on its own platform. A customised application can be viewed on a device to see how it looks, before it is made live.

The pros:

1. Provides a GUI based build system
2. Supports template projects to help kick-start hybrid application development
3. Continuous support of Adobe and hundreds of open source developers

The cons:

1. Lacks strong performance with graphic intensive mobile applications
2. At times, it shows a lot of variation in performance, which results in it being unstable

Appcelerator

Appcelerator is a platform that is compatible with Android, Windows and iOS. It has the drag-and-drop feature for easy placement of different objects. It also has the Hyperloop feature, which helps in the use of JavaScript to get direct access to different native APIs present in iOS and Android. Another good feature available with this platform is that it provides real-time performance and crash analytics. This enables the finding and fixing of issues within the application itself. It supports the development of native mobile, tablet as well as desktop applications using different Web programming languages like PHP, HTML, JavaScript,



Figure 2: Top six hybrid mobile app development frameworks (source: googleimages.com)



Figure 3: PhoneGap development (source: googleimages.com)